

EDUCATION

Nanyang Technological University, Singapore Aug 2025 – May 2029

Bachelor of Science in Mathematical and Computer Sciences

- **Honours:** 4.98 CGPA, Dean's List for Year 1 (Under both mathematics and computing schools)
- **Courses:** Linear Algebra II (A+), Calculus II (A+), Algorithms & Data Structures (A+)

Philippine Science High School - Main Campus (PSHS-MC) Aug 2019 – May 2025

High School Diploma in STEM

- **Honours:** Highest Honors, Best in Math and CS Awards

HONOURS AND AWARDS

International Collegiate Programming Contest 2025, Manila Regional (Bronze, 7th) Dec 2025 – Dec 2025

- Attained 7th place in collaboration with two teammates on C++ programming challenges, achieving bronze

International Olympiad in Informatics (IOI) 2025 (Honorable Mention) Jul 2025 – Aug 2025

- Devised C++ solutions for ad-hoc programming challenges in five hours, attaining an Honorable Mention

Asia-Pacific International Olympiad (APIO) 2025 (Bronze Medalist) May 2025 – May 2025

- Utilised skills in olympiad-level combinatorics to solve nonstandard problems with C++, earning a bronze medal

Philippine Physics Olympiad 2025 (Bronze Medalist) Mar 2025 – Mar 2025

Singapore International Mathematical and Computational Challenge 2024 May 2024 – May 2024

- Co-authored a technical machine learning paper within a 24-hour time constraint on sparse signal detection of 100000 25px by 25px grayscale images with a k-means clusterer using an entropy maximization heuristic with Python, Sklearn, and Numpy, placing top 10 and receiving a special award in creativity

ACADEMIC PROJECTS

Nanyang Technological University, Singapore Sep 2026 – Sep 2026

Algorithm Design and Analysis Project - Runtime Analysis and Hyperparameter Tuning of Hybrid Sort

- Modeled runtime behavior of hybrid sort and applied unconstrained continuous optimization techniques, resulting in a general theoretical model for optimizing runtime of hybrid sort
- Performed a two-variable regression analysis using a custom gradient descent pipeline in Python and TensorFlow inspired by machine learning techniques to compute optimal hybrid sort hyperparameter values over a dataset of empirical runtime data, decreasing sorting time by 7.6% for arrays with 10 million elements
- Wrote a 10-slide strong induction proof of the optimality of some set of hyperparameter values, providing theoretical airtightness in the analysis

Nanyang Technological University, Singapore Oct 2025 – Nov 2025

Introduction to Computational Thinking & Programming Project - Simulated Annealing Team Allocator

- Reformulated diverse team allocation as a discrete computational optimization problem over a constructed feature space and applied Simulated Annealing in Python on 6000 rows of student educational records to achieve groupings with near-mathematically-optimal diversity
- Proved that the formulated optimization problem is in NP-hard (nondeterministic polynomial time—hard), rigorously supporting the use of heuristic methodologies

PERSONAL PROJECTS

Nanyang Technological University, Singapore Nov 2025 – Nov 2025

Verilog Computer

- Programmed a computer using Icarus Verilog and designed a custom assembly language and assembler, solidifying my understanding of the concepts presented in NTU's Digital Logic course

Generative Adversarial Network for Generating Anime Character Faces Jun 2022 – Jul 2022

- Researched and implemented Generative Adversarial Networks (GANs) using Python with Tensorflow over the summer, improving my research and self-study skills

Planks and Flanks

Apr 2021 – Apr 2021

- Developed a full-stack multiplayer game with JavaScript, Node.JS, and Socket.io within two days, enhancing my ability to integrate algorithms knowledge with full-stack development

WORK EXPERIENCE

National Olympiad in Informatics, Philippines (NOI.PH)

May 2025 – Aug 2025

International Olympiad in Informatics (IOI) Training Director

- Directed a 3-month long training program for the Philippine IOI team, curating problem sets covering dynamic programming, data structures, and graph algorithms, and providing technical mentorship and moral support, leading to a historic first IOI gold medal

Achieve Without Borders, Inc.

Jun 2023 – Jul 2023

Software Development Intern

- Developed a full-stack proof-of-concept software application with Dart, Flask, and Odoo within two weeks, refining my ability to efficiently learn a new framework within a limited timeframe

LEADERSHIP AND CO-CURRICULAR ACTIVITIES

Pisay Competitive Programming Competition (PCPC)

Jun 2023 – May 2025

Founding Overall Head (2023-2024), Overall Co-head (2024-2025)

- Led first iteration of a national team-based programming contest for Filipino high-schoolers, increasing participation from 23 to 75 students while orchestrating a team of 15-20 students

Reboot

Jun 2023 – May 2025

Overall Co-head (2023-2024), Overall Head (2024-2025)

- Directed a school-wide competitive programming training pool, contributing to 8 students placing in top 20 of the 2025 NOI.PH National Eliminations

LICENSES AND CERTIFICATIONS

Google AI Professional Certificate

Aug 2025 – Aug 2025

Issued by *Coursera*

ID: "https://www.credly.com/go/r1A05A6j"

SKILLS, LANGUAGES, INTERESTS

- **Languages:** English (Native speaker), Filipino/Tagalog (Native speaker)
- **Programming Languages:** C++, Python, Dart, JavaScript
- **Libraries and Frameworks:** Sklearn, Numpy, Tensorflow, Flask, Node.JS
- **Interests:** Minecraft city planning and transit design, anime, rhythm gaming, train enthusiast, improvisational piano, blog writing, karaoke, memorizing maps and transit diagrams